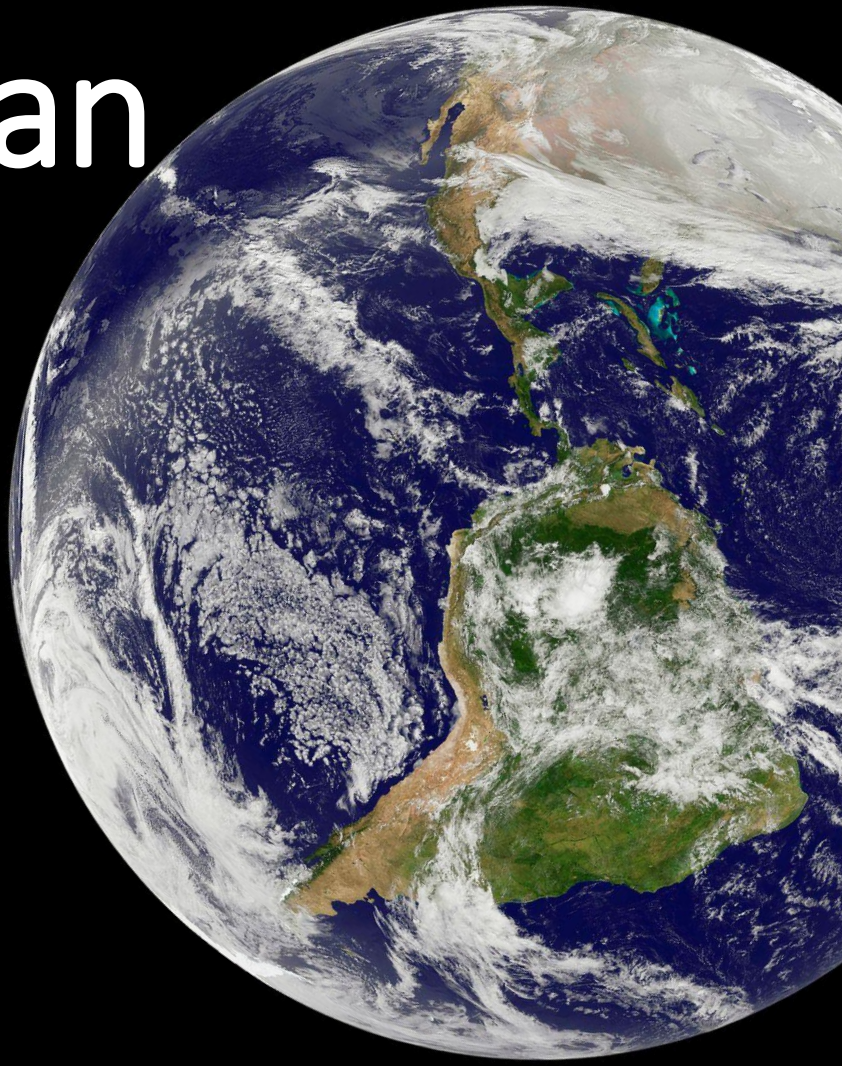
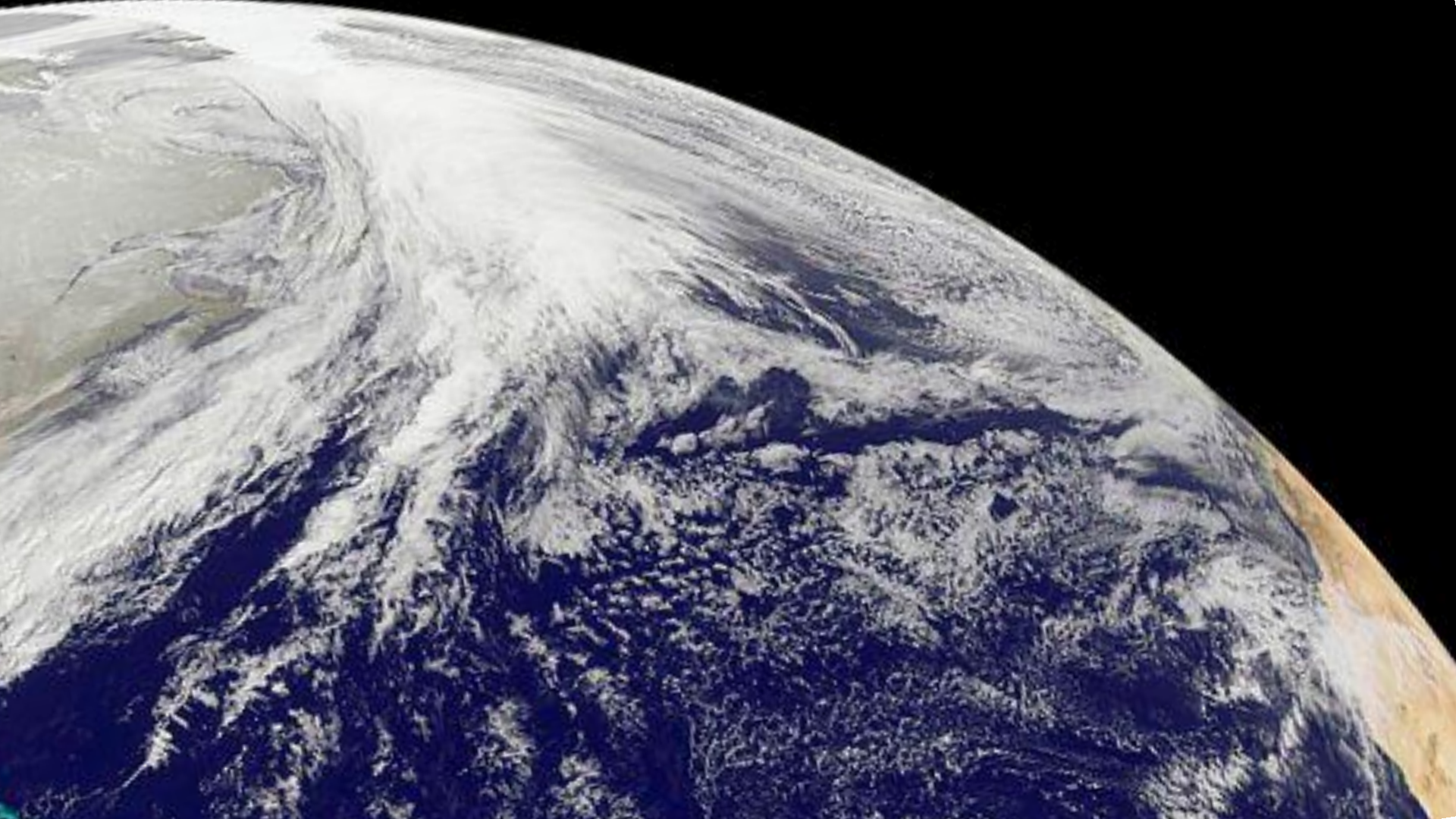
Presentation Plan

- Introduction
- Background & Literature Review
- Methodology
- Results
- Discussion
- Stakeholders: Interests, Conflicts & Solutions
- Conclusions

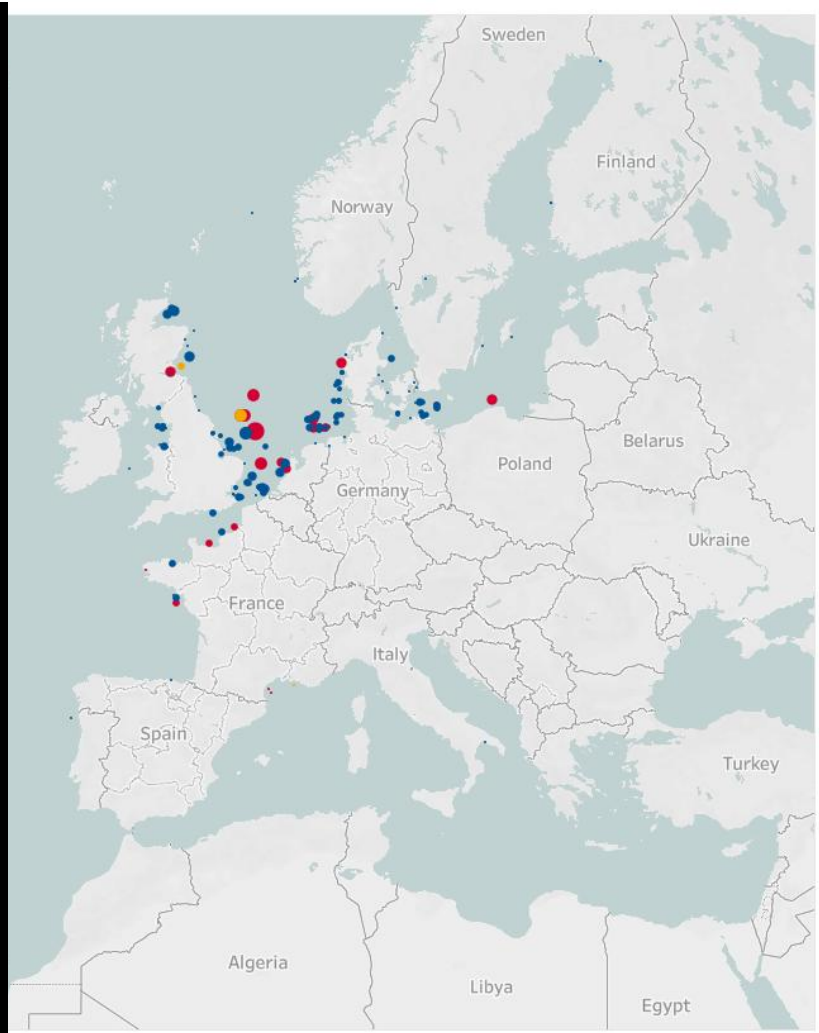




Rising electricity demand & climate goals

- **Boost renewables & offshore wind:** 45% renewables by 2030; 60 GW offshore wind by 2030, 300 GW by 2050.
- **Stabilize prices & unlock investment**
- **Grid upgrades & EU goals:** Smart interconnectors to meet Paris Agreement & EU Green Deal targets.

- **Offshore wind: scalable, low-carbon, fewer land-use conflicts**



Europe's Offshore Wind
36,657 MW connected to the grid
6,592 turbines connected
139 Wind Farms
13 countries

Status
- Online
- Partially online
- Under construction

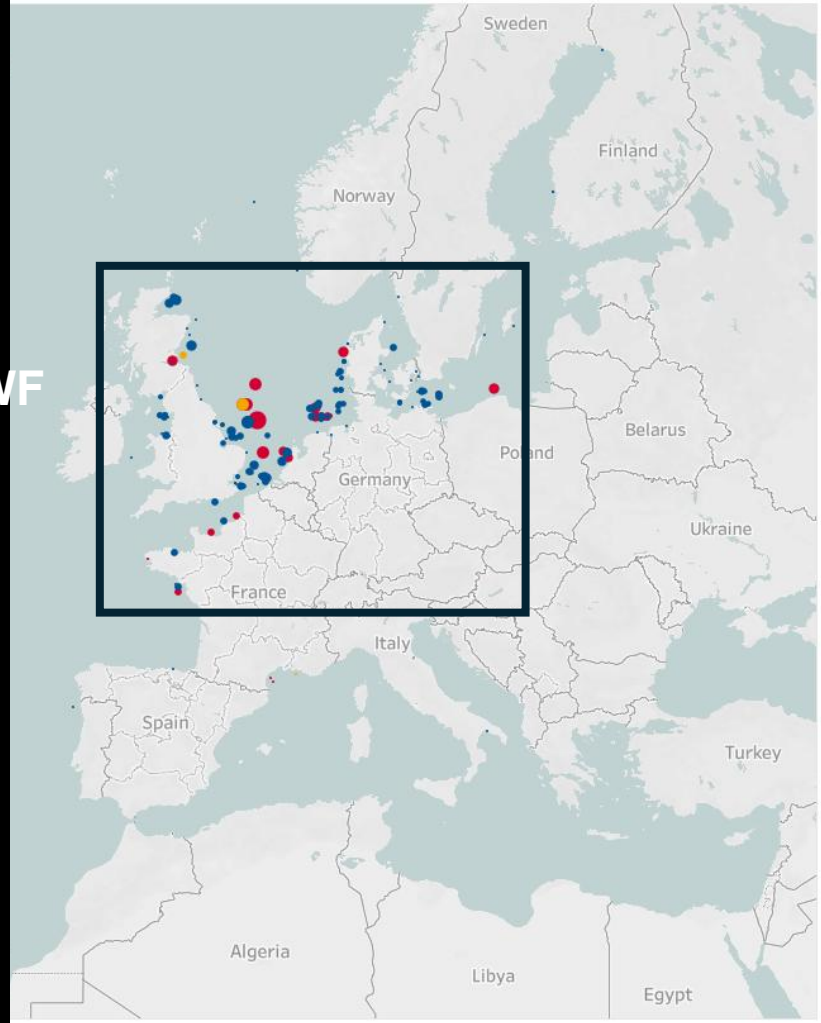
Nature
(All)

	MW Connected	Turbines Connected	Number of projects
United Kin..	15,932	2,861	47
Germany	9,121	1,639	33
Netherlands	4,739	670	10
Denmark	2,653	672	17
Belgium	2,261	399	11
France	1,500	217	5
Sweden	192	80	5
Norway	101	13	3
Finland	71	19	3
Italy	30	10	1
Ireland	25	7	1
Portugal	25	3	1
Spain	7	2	2

Last update: August 6, 2025

Share

- **Offshore wind: scalable, low-carbon, fewer land-use conflicts**
- **Northern Europe leads OWF UK, Germany, Denmark**
- **EU targets: 60 GW by 2030, 300 GW by 2050**



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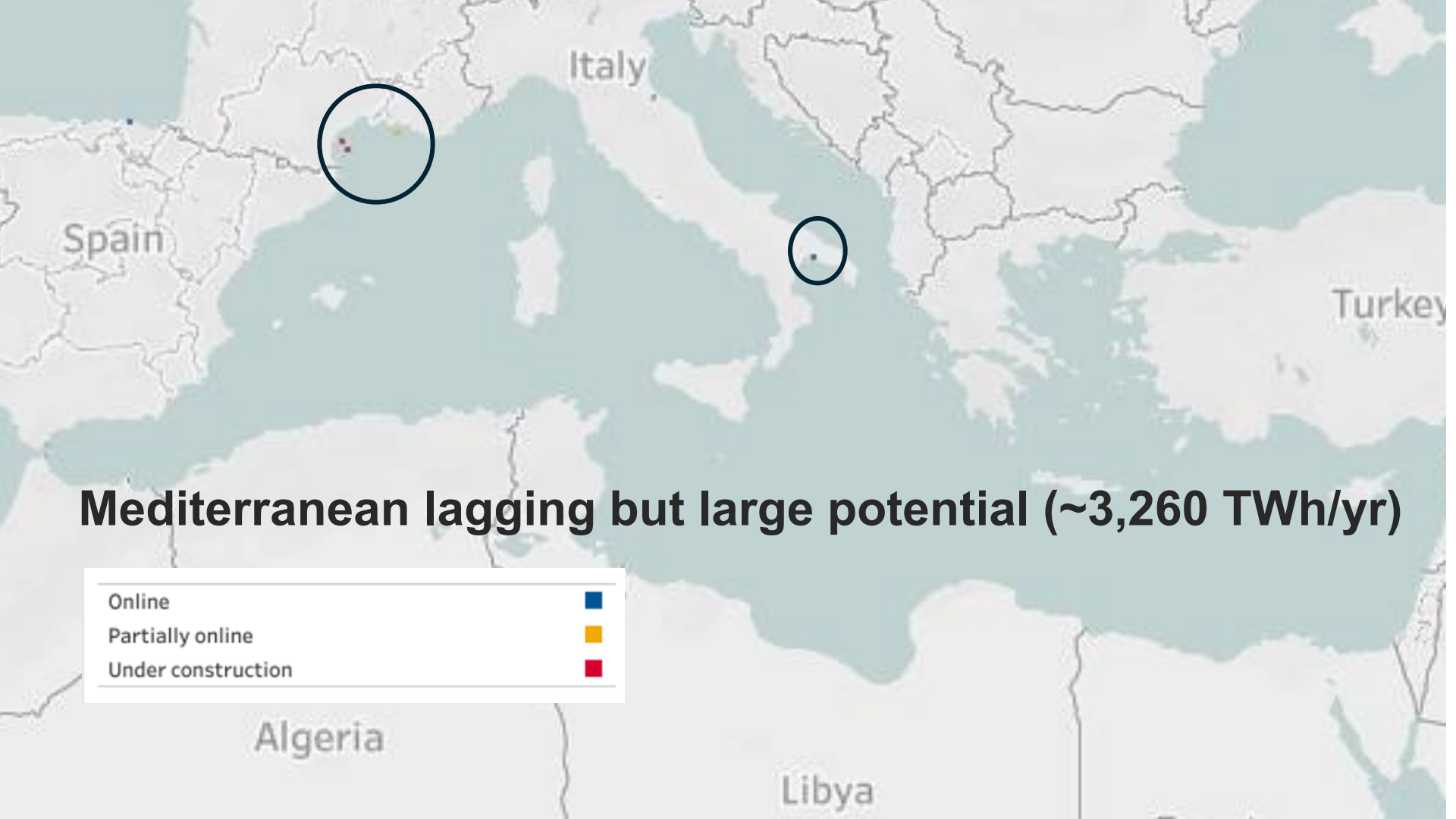
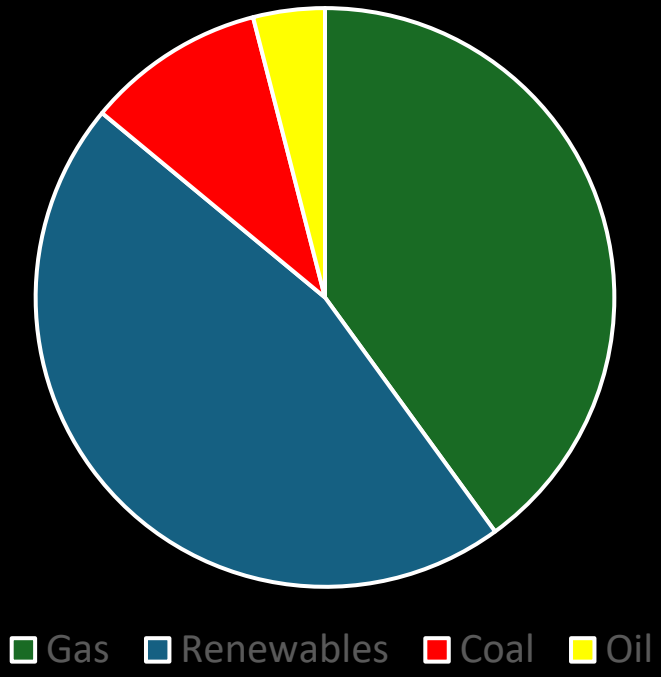


Mediterranean lagging but large potential (~3,260 TWh/yr)

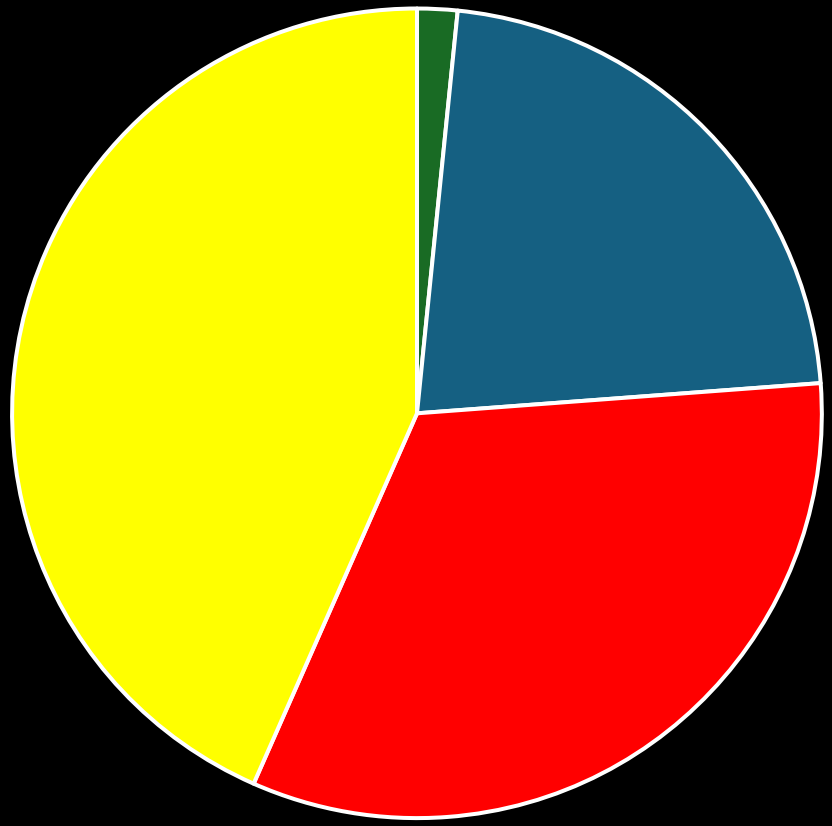
- **NECP Targets (National Energy & Climate Plan):**
  - *Installed renewable capacity*: **25 GW** by 2030 (from ~11 GW today).
  - *Renewable electricity share*: **80% by 2030**.
  - *Coal phase-out*: All lignite plants to close by **2028**.
  - **1.9 GW** *OWF by 2030*
- **EU Compliance:**
  - Aligned with **EU Green Deal** & **Paris Agreement** (Net Zero 2050).
  - Implements **EU Fit for 55** package (−55% GHG by 2030).

Gas ~40%,
Renewables ~46%,
Coal ~10% (declining),
Oil ~4% (islands)

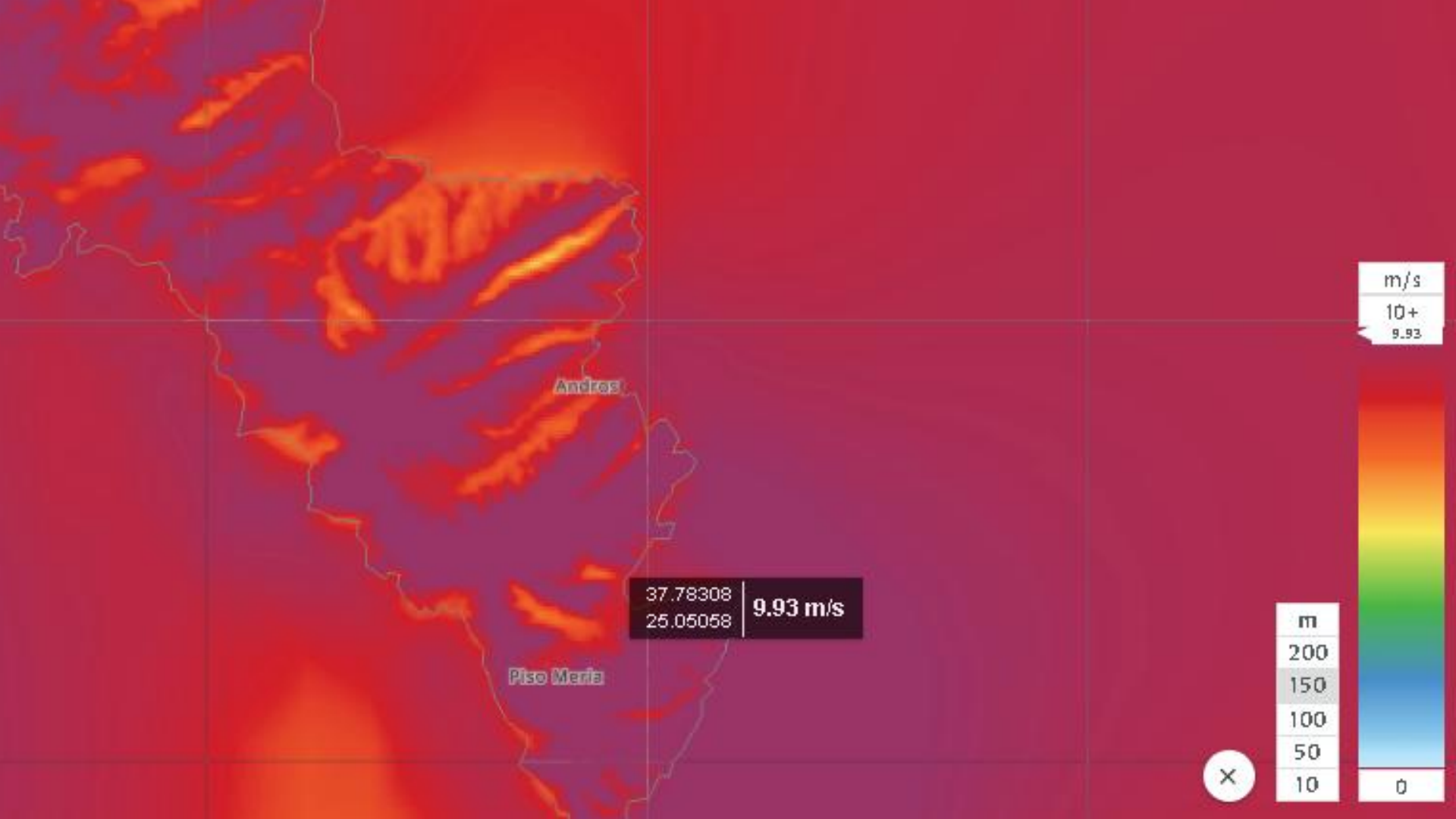
Energy Mix



Renewables



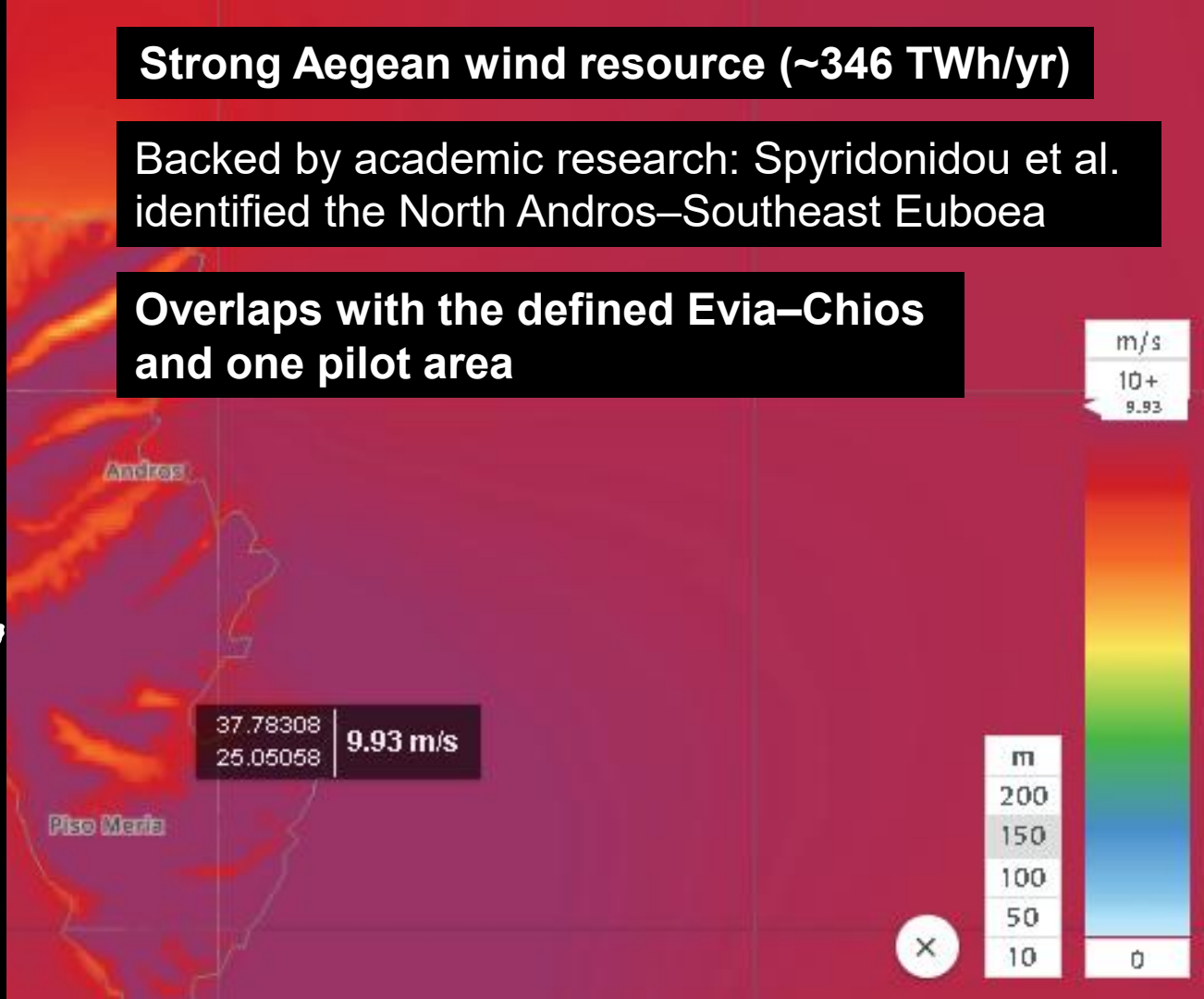
LOCATION	CAPACITY
Evros and Samothraki	600 MW
Crete Island	600 MW
Cyclades (Syros, Paros, Naxos)	250 MW
Cycladic Islands (Attica, S. Evia)	200 MW
Dodecanese Islands	350 MW


Andros
Piso Meria
37.78308
25.05058
9.93 m/s
m/s
10+
9.93
0
m
200
150
100
50
10

Strong Aegean wind resource (~346 TWh/yr)

Backed by academic research: Spyridonidou et al. identified the North Andros–Southeast Euboea

Overlaps with the defined Evia–Chios and one pilot area



Assessment of Potential for Offshore Wind Power Generation in Greece

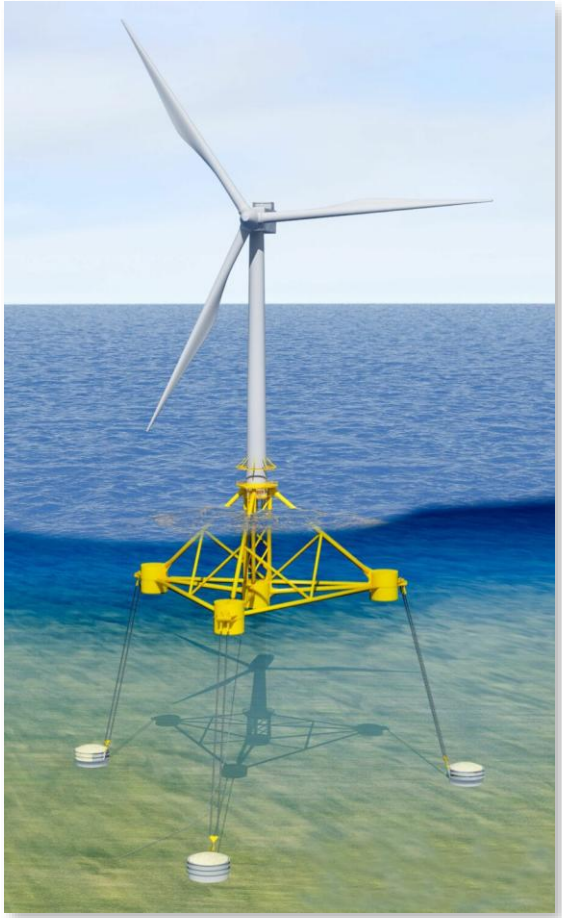
Objective: ***Techno-economic feasibility of a 1.9 GW floating OWF in SE Euboea–North Andros, Greece***

Oliwier Sobczyk, supervisor Dr. Viney Dixit

Technical Analysis Framework

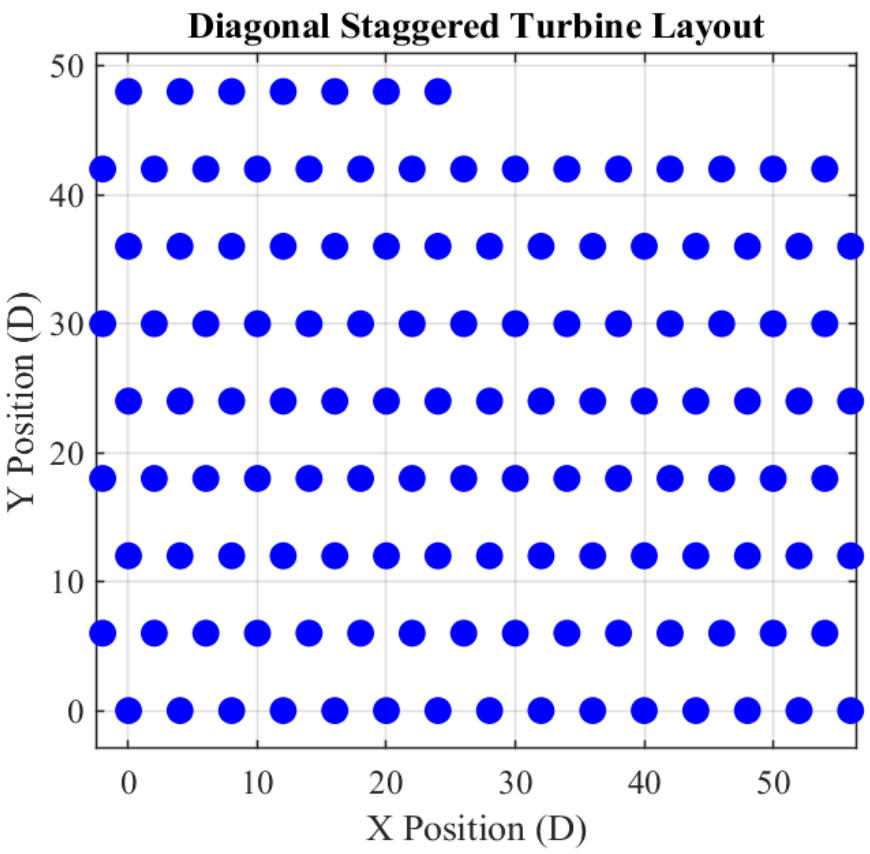
by Schütte – parallel study

- Turbine: Vestas V236–15 MW (127 units)
- Foundations: Floating semi-submersible INO15C
- Export cable: HVAC, ~47 km to shore
- Hexagonal micro-siting
- Port distance ~90 km



Results: Farm Layout

- 127 × Vestas V236–15 MW
- Hexagonal array · 148 km2
- Installed capacity: 1,905 MW



Methodology Overview

- MATLAB-based Jensen/PARK wake model
- Data: ERA5 wind fields (100 m)
- Economics: CAPEX, OPEX, DECOM, NPV, IRR, LCOE

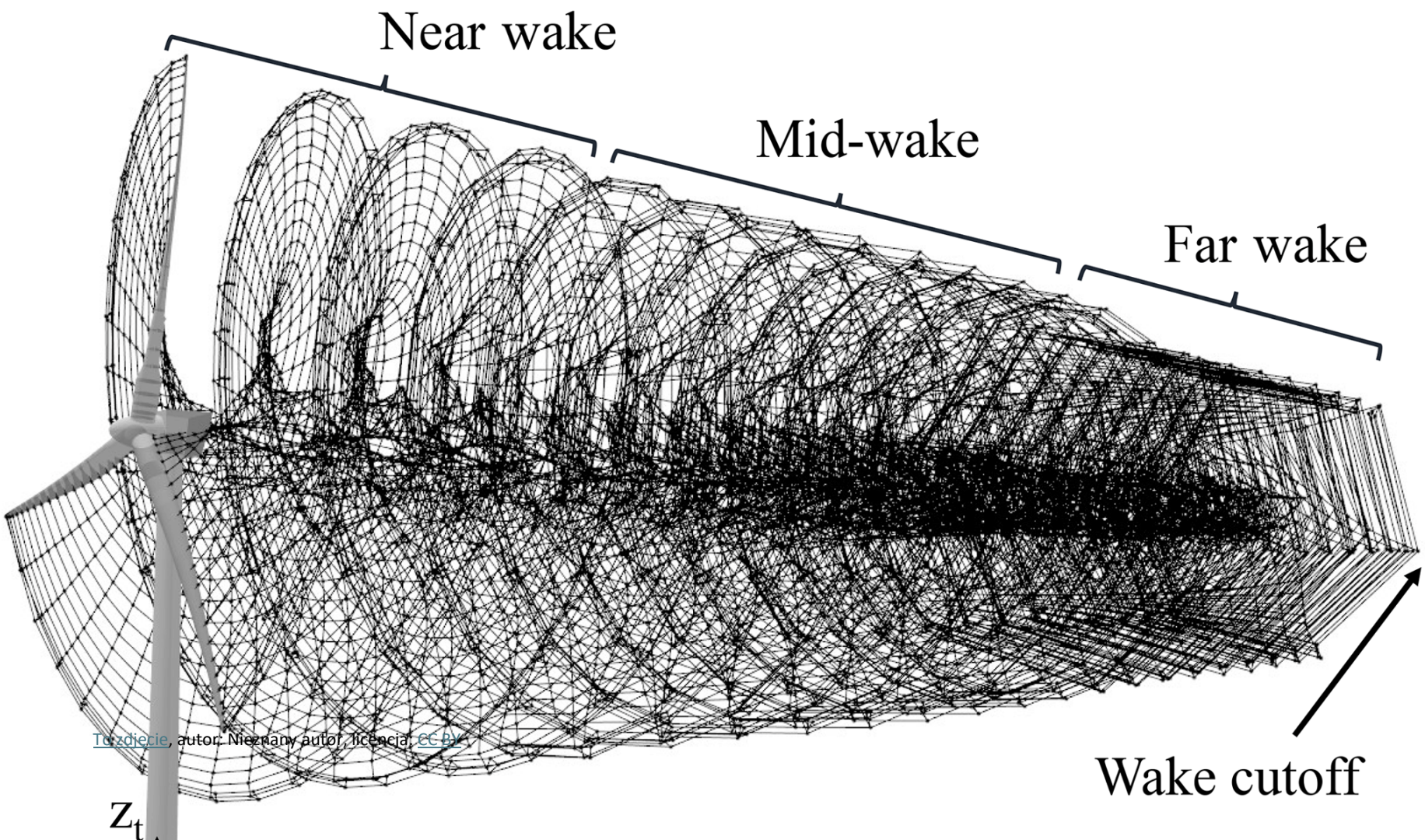
Methodology: Wind Field Reconstruction

$$U_i^{(d)}(t) = \sqrt{u_i(t)^2 + v_i(t)^2}. \qquad (1)$$

$$U_i(t) \;=\; U_i^{(d)}(t)\left(\frac{z_h}{z_d}\right)^{\alpha}, \qquad (2)$$

$$\theta(t) = \mathrm{mod}\left(\mathrm{arctan2}\left(\overline{\sin\theta_i(t)}, \overline{\cos\theta_i(t)}\right), 2\pi\right), \qquad (3)$$

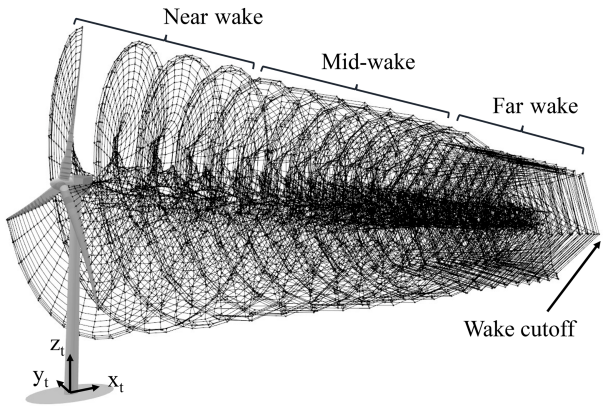
$$\rho(t) \;=\; \rho_0 \frac{T_0}{T(t)} \;=\; 1.208 \frac{288.15\ K}{T(t) + 273.15\ K}, \qquad (4)$$



To zdjęcie, autor: Nieznany autor, licencja: CC BY.

Methodology: Wake Modelling

- Jensen/PARK top-hat model
- Wake radius: R = D/2 + k·x (Eq. 7)
- k from turbulence intensity (Eq. 8–9)
- Deficits combined by RSS; effective inflow U_eff (Eq. 10–12)



$$\delta_{ij}(t) = \left(1-\sqrt{1-C_T(t)}\right)\left(\frac{D}{D+2\,k(t)\,x_{ij}(t)}\right)^2, \tag{10}$$

$$\delta_i(t) = \sqrt{\sum_{j\in\mathcal{S}_i(t)} \left(\phi_{ij}(t)\,\delta_{ij}(t)\right)^2}, \tag{11}$$

$$U_i^{\text{eff}}(t) = U_i(t)\left(1-\delta_i(t)\right). \tag{12}$$

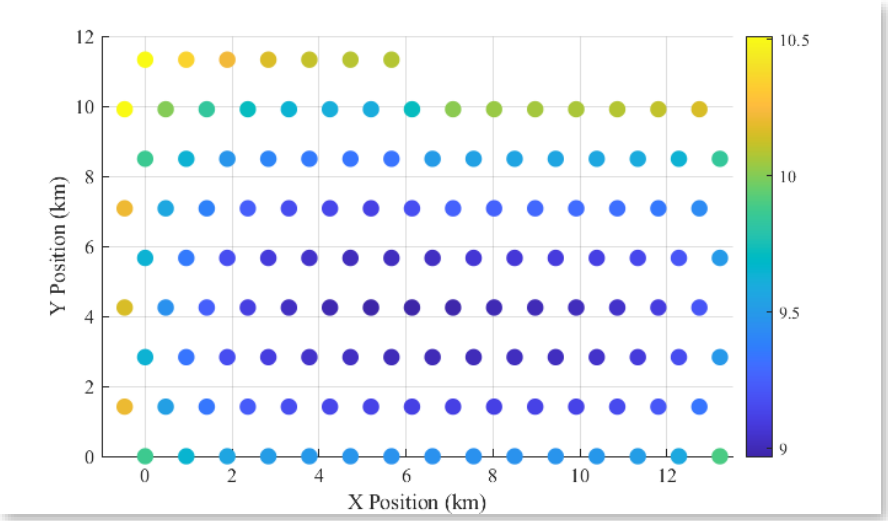
$$R_j(x,t) = \frac{D}{2} + k(t)\,x, \qquad x \geq 0, \tag{7}$$

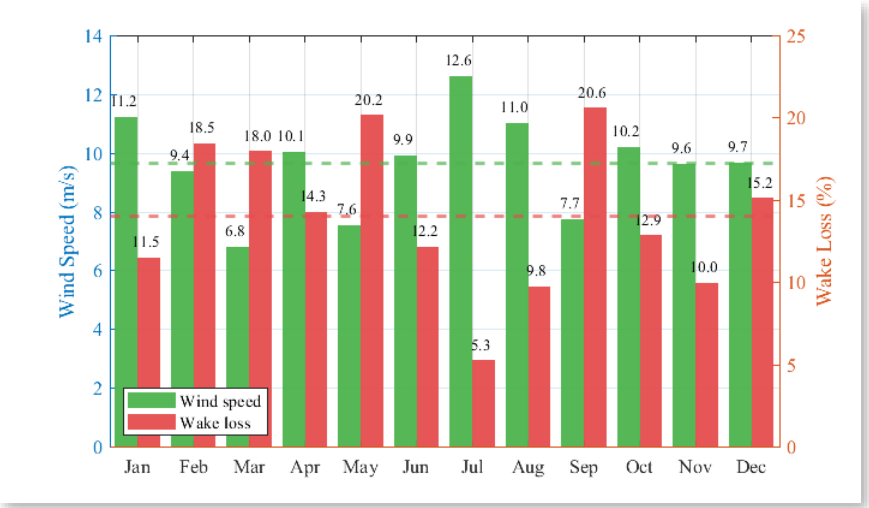
$$k(t) = 0.38\,I_0^{\text{eff}}(t) + 0.0037 \tag{8}$$

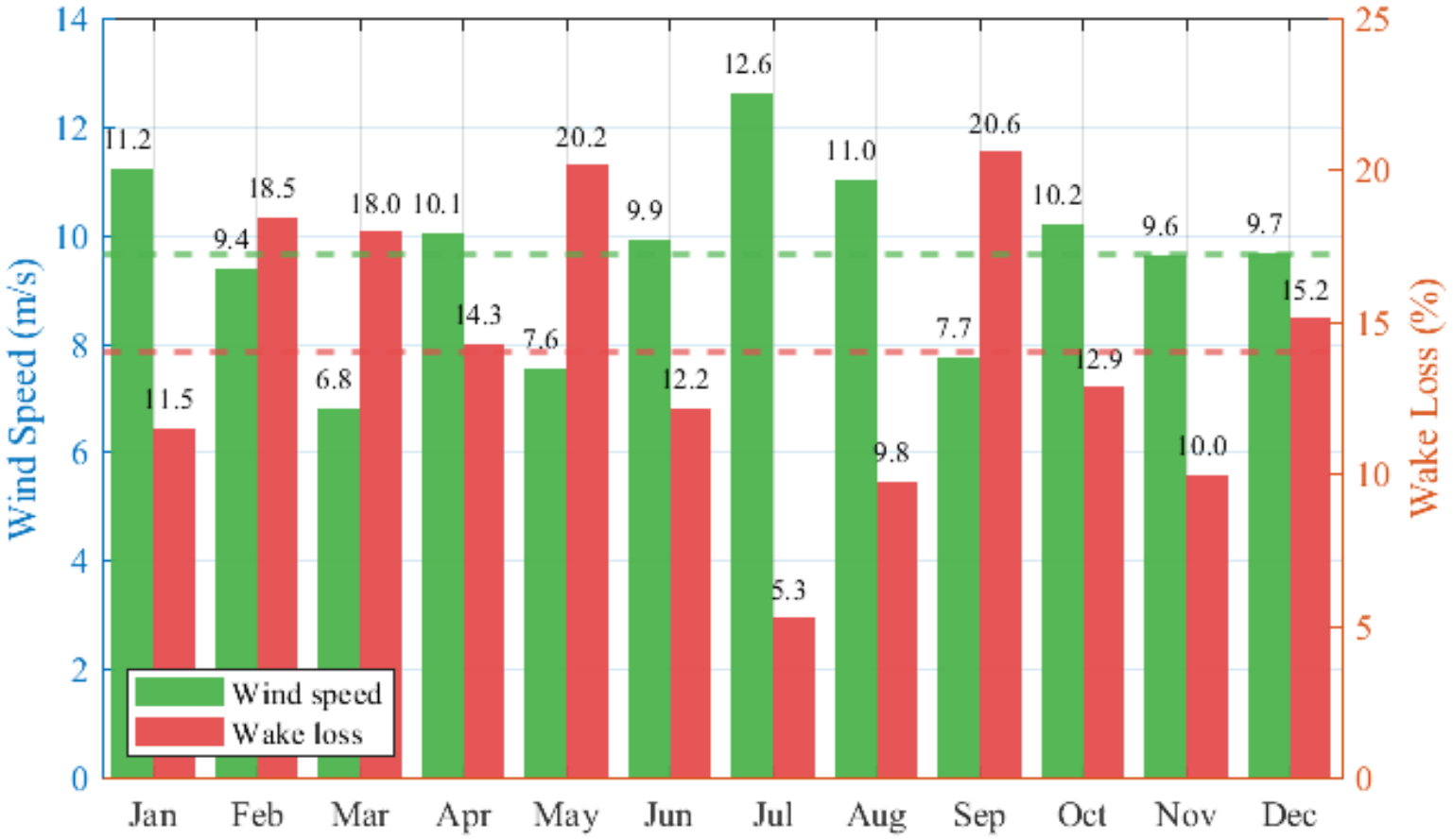
$$I_0^{\text{eff}}(t) = \tfrac{1}{2}\sqrt{I_0^2 + \left(\frac{\sigma_i^u(t)}{U_i(t)}\right)^2} \tag{9}$$

Results: Wake Impacts

- Average wake losses: 14% (range 5–21% monthly)
- Stronger in sub-rated Region 2; weaker at rated (CTˇ)






Wind Speed (m/s)
Wake Loss (%)
0
2
4
6
8
10
12
14
0
5
10
15
20
25
11.2
11.5
9.4
18.5
6.8
18.0
10.1
14.3
7.6
20.2
9.9
12.2
12.6
5.3
11.0
9.8
7.7
20.6
10.2
12.9
9.6
10.0
9.7
15.2
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Wind speed
Wake loss

Methodology: Power & Thrust Curves

$$\rho(t) = \rho_0 \frac{T_0}{T(t)} = 1.208 \frac{288.15\ K}{T(t) + 273.15\ K}, \quad (4)$$

$$U_i^{\text{eff}}(t) = U_i(t)\left(1 - \delta_i(t)\right). \quad (12)$$

$$P(U_i, \rho) = \begin{cases} 0, & U_i < v_{\text{ci}} \text{ or } U_i > v_{\text{co}}, \\ \dfrac{P_r}{1 + e^{-(b U_i - 7.5)}} \dfrac{\rho}{\rho_0}, & v_{\text{ci}} \leq U_i \leq v_r, \\ P_r \dfrac{\rho}{\rho_0}, & v_r < U_i \leq v_{\text{co}}, \end{cases}$$

$$\text{with } b = 5.822\, e^{-0.3398\, v_r} + 1.79\, e^{-0.0548\, v_r}. \quad (5)$$

Methodology: Power & Thrust Curves

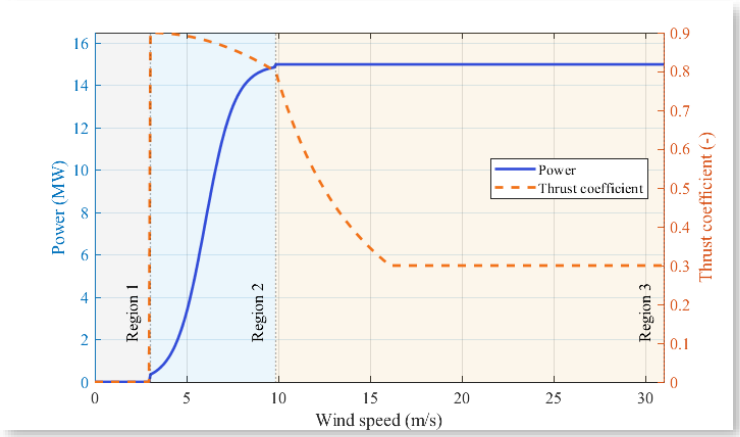
equation $P = \frac{1}{2}\,\rho\,A\,\Delta V^3\,C_p$

$$P(U_i,\rho) = \begin{cases} 0, & U_i < v_{\mathrm{ci}} \text{ or } U_i > v_{\mathrm{co}}, \\ \dfrac{P_r}{1+e^{-(bU_i-7.5)}}\,\dfrac{\rho}{\rho_0}, & v_{\mathrm{ci}} \le U_i \le v_r, \\ P_r\,\dfrac{\rho}{\rho_0}, & v_r < U_i \le v_{\mathrm{co}}, \end{cases}$$

$$\text{with } b = 5.822\,e^{-0.3398\,v_r} + 1.79\,e^{-0.0548\,v_r}. \qquad (5)$$

$$C_T = \begin{cases} 0, & U_i < v_{\mathrm{ci}}\,,\, U_i > v_{\mathrm{co}}, \\ 0.9 - 0.1\left(\frac{U_i - v_{\mathrm{ci}}}{v_r - v_{\mathrm{ci}}}\right)^2, & v_{\mathrm{ci}} \le U_i \le v_r, \\ \max\left(0.3,\, 0.8\left(\frac{v_r}{U_i}\right)^2\right), & v_r < U_i \le v_{\mathrm{co}}. \end{cases} \qquad (6)$$

$$P_\Sigma(t) = \sum_{i=1}^{N} P\left(U_i^{\mathrm{eff}}(t), \rho(t)\right). \qquad (13)$$



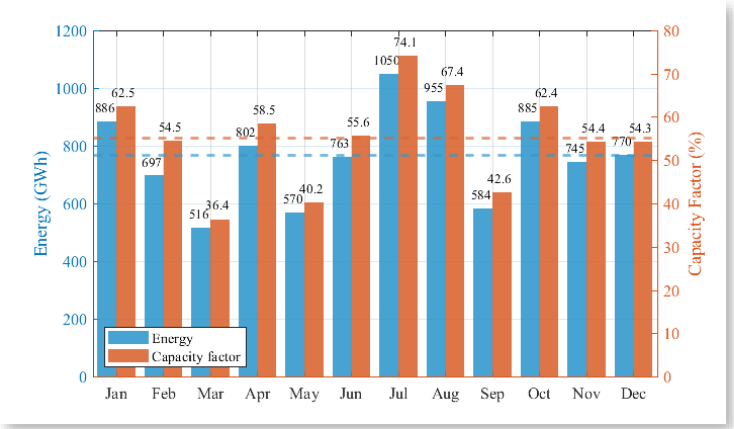
- Cut-in 3 m/s · Rated 9.81 m/s · Cut-out 31 m/s
- Power curve uses exponential model (Eq. 5)
- Thrust coefficient curve (Eq. 6) drives wakes

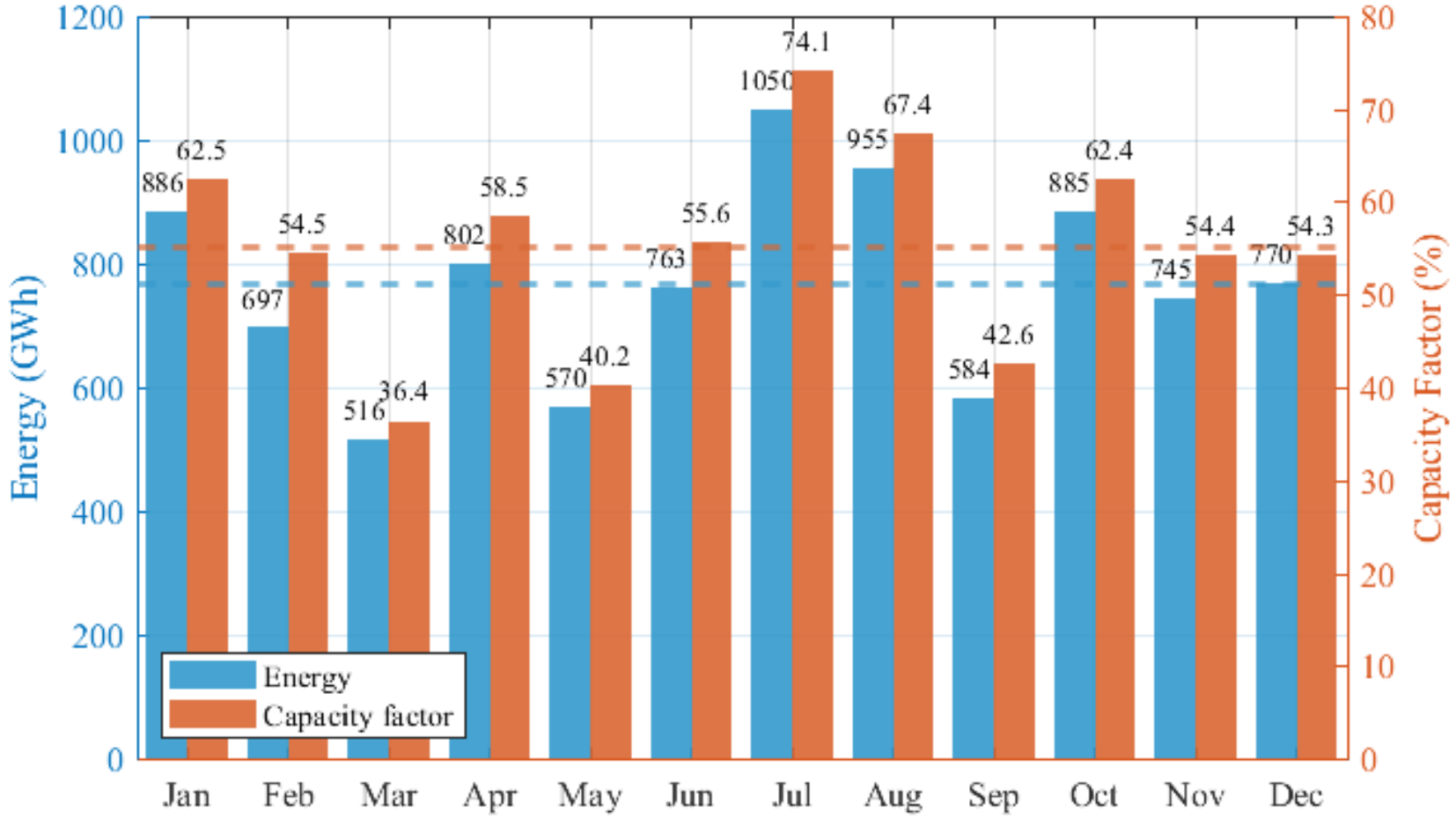
$$E_{\mathcal{T}} = \sum_{t\in\mathcal{T}} P_{\mathrm{net}}(t)\,\Delta t, \qquad \Delta t = 1\ \mathrm{h}. \qquad (15)$$

$$P_{\mathrm{net}} = P_\Sigma\,(1 - L_{\mathrm{elec}})\,(1 - L_{\mathrm{array}})\,A, \qquad (14)$$

Results: Energy Production

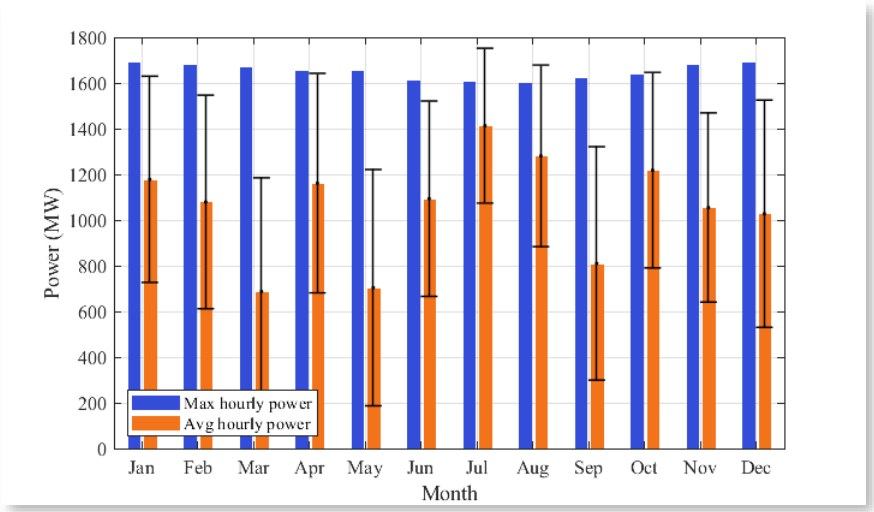
- Annual Energy Production (AEP): 9,216 GWh
- Average power: 1054 MW
- Capacity factor: 55.3%
- Average Monthly Production: 768 GWh

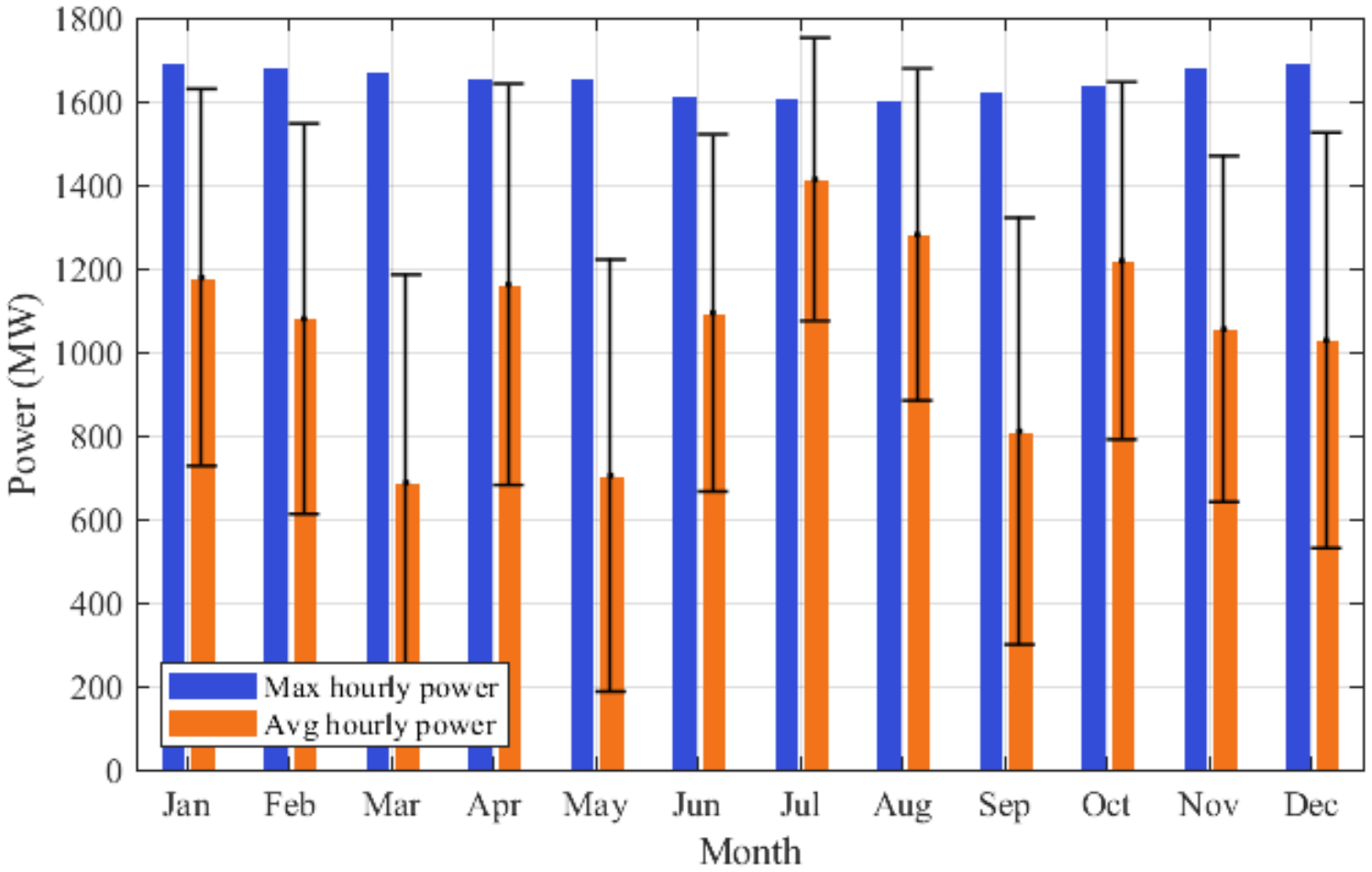



Energy (GWh)
Capacity Factor (%)
1200
1000
800
600
400
200
0
80
70
60
50
40
30
20
10
0
886
62.5
697
54.5
516
36.4
802
58.5
570
40.2
763
55.6
1050
74.1
955
67.4
584
42.6
885
62.4
745
54.4
770
54.3
Energy
Capacity factor
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec

Results: Power Output Stability (Monthly)

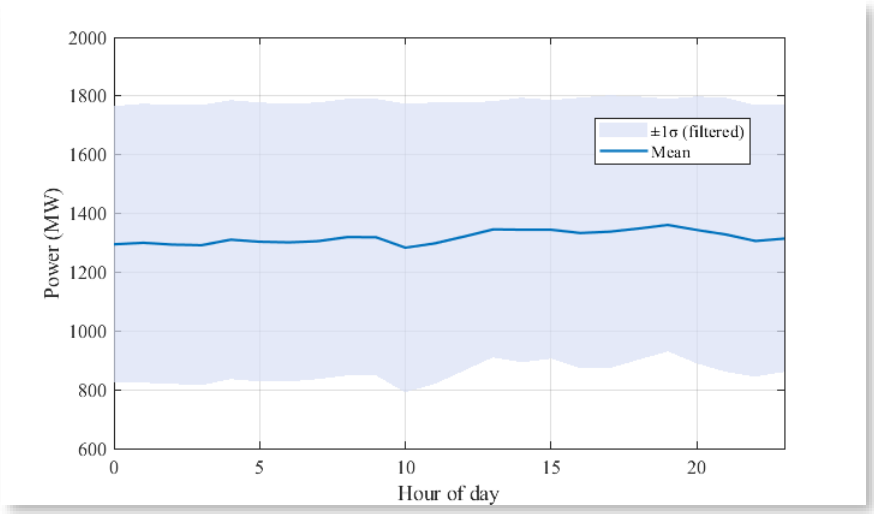
- Seasonal variation: ~500–1,050 GWh/month
- Std dev: 161 GWh (~20%)

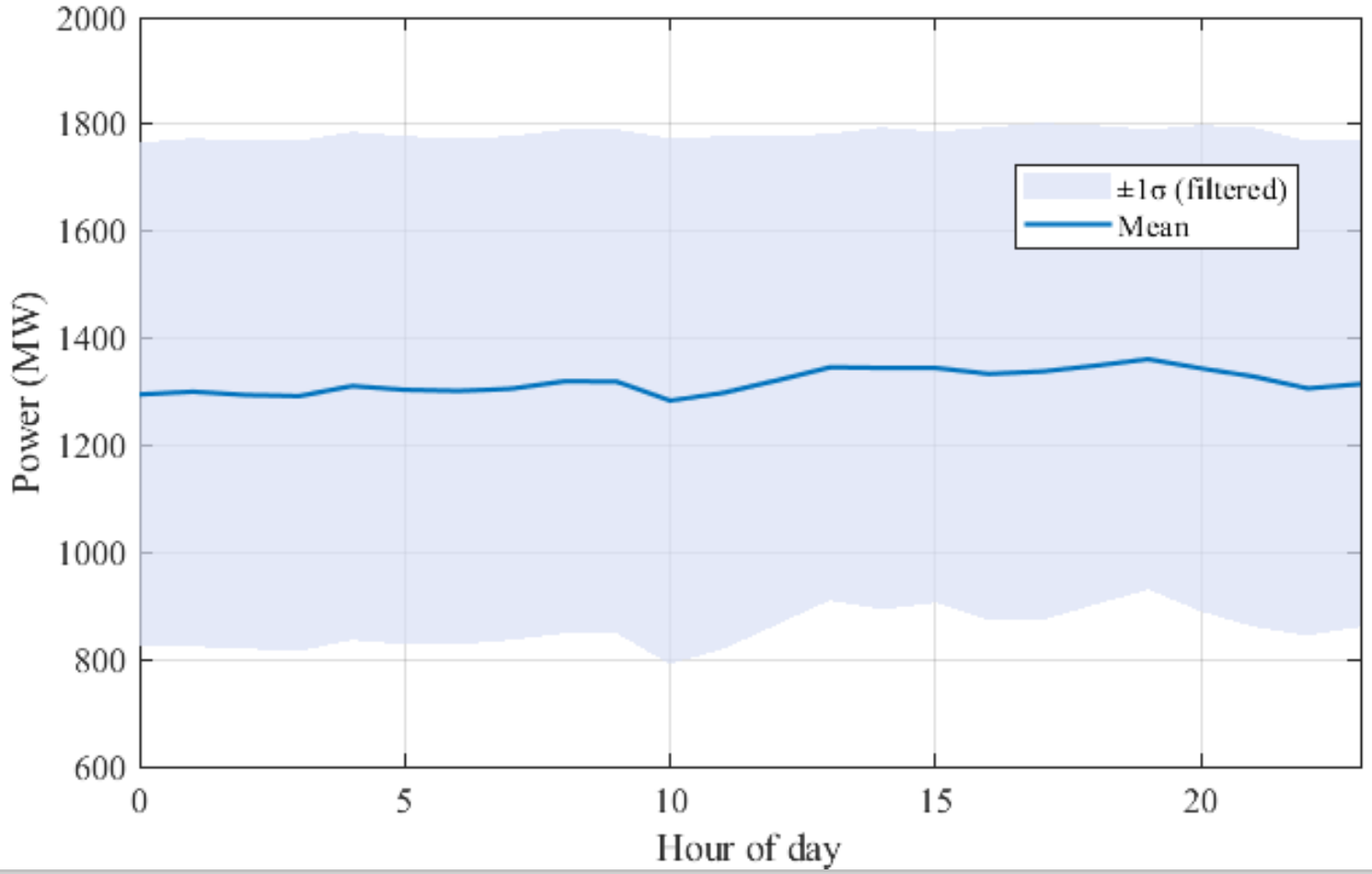



1800
1600
1400
1200
1000
800
600
400
200
0
Power (MW)
Max hourly power
Avg hourly power
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Month

Results: Power Output Stability (Daily)

- Daily average 1,200–1,400 MW
- Standard deviation 463 MW (filtered)




2000
1800
1600
1400
1200
1000
800
600
Power (MW)
0
5
10
15
20
Hour of day
±1σ (filtered)
Mean

Methodology: Economic Model

- CAPEX = €4.7m/MW › ~€8.95bn (1.905 GW)
- OPEX = 3–4% of CAPEX › ~€313m/yr
- DECOM = 8% of CAPEX (discounted)
- NPV = PVR - PVC (Eq. 21)
- IRR -internal rate of return : solve NPV(r)=0
- LCOE (Eq. 22)
- r = 9%

$$\mathrm{PV}_R = \sum_{y=1}^{T} \frac{R_y}{(1+r)^y}, \quad R_y = E_y p_y$$

$$\mathrm{PV}_C = \mathrm{CAPEX} + \sum_{y=1}^{T} \frac{\mathrm{OPEX}_y}{(1+r)^y} + \frac{\mathrm{Decom}}{(1+r)^T}. \tag{19}$$

$$\mathrm{CAPEX} = C_{\mathrm{tot}} \times \mathrm{CAPEX}_{\mathrm{per\ MW}}. \tag{16}$$

$$\mathrm{OPEX}_y = C_{\mathrm{tot}} \times \mathrm{OPEX}^{\mathrm{fix}}_{y\mathrm{perMW}} \tag{17}$$

$$\mathrm{DECOM} = C_{\mathrm{tot}} \times \mathrm{DECOM}^{\mathrm{fix}}_{\mathrm{perMW}} \times \frac{1}{(1+r)^T} \tag{18}$$

$$p_y = p_0 (1+g)^{y}, \tag{20}$$

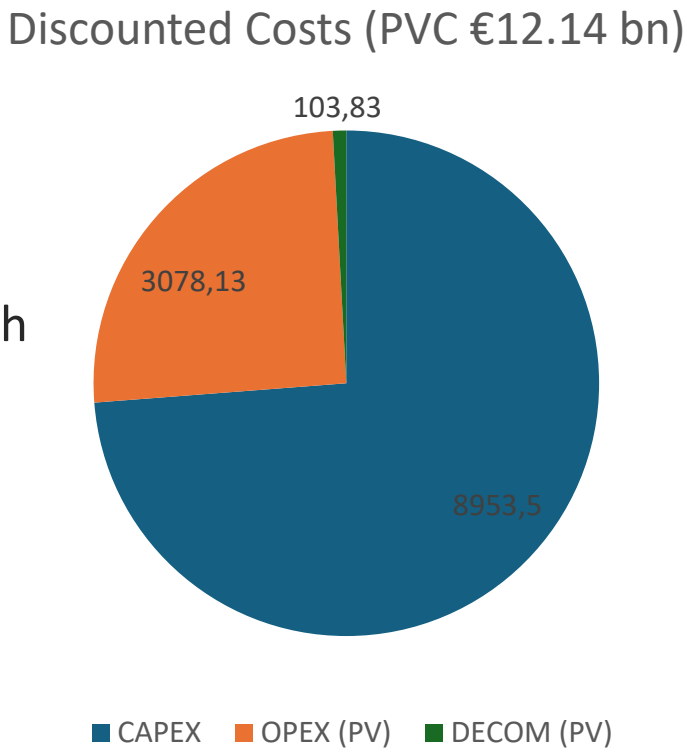
$$g = -d_{\mathrm{WF}} \left(\frac{P_{\mathrm{net}}}{P_{\mathrm{GrTot}}} \right) + d_{\mathrm{infl}},$$

$$\mathrm{NPV} = \mathrm{PV}_R - \mathrm{PV}_C. \tag{21}$$

$$\mathrm{LCOE} = \mathrm{PV}_C \left(\sum_{y=1}^{T} \frac{E_{\mathrm{year}}}{(1+r)^y} \right)^{-1}. \tag{22}$$

Economic Performance Summary

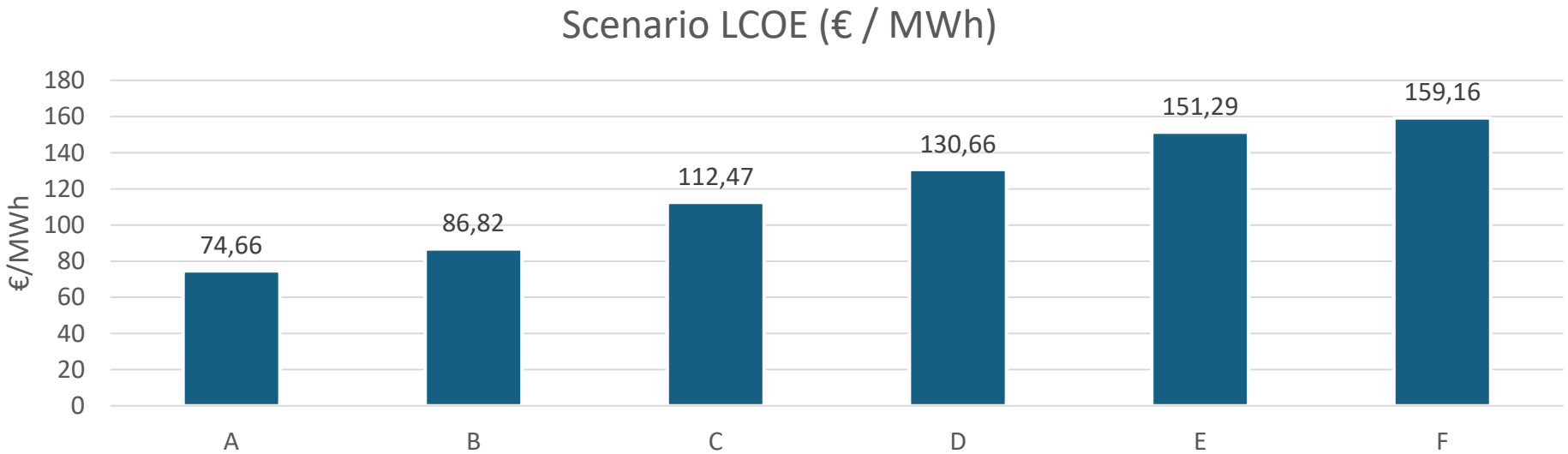
- NPV: €5.46 bn;
- IRR: 17.3%;
- Payback: ~6.7 years
- LCOE: €134/MWh [€75/MWh €159/MWh
- €7 Billion Funding from Greek Gov.
- EU total budget: € 672.5 billion



Sensitivity Analysis (LCOE)

- Base case: €134/MWh
- Best case: €75/MWh (-44%)
- Worst case: €159/MWh (+19%)
- Main drivers: CAPEX ˇ and AEP ^

Case	δC (%)	δO (%)	δE (%)	LCOE	Change (%)
A	-40	-10	+5	74.66	-44.31
B	-40	+10	-5	86.82	-35.24
C	-10	+10	+10	112.47	-16.11
D	+10	-10	+10	130.66	-2.54
E	+10	-10	-5	151.29	+12.85
F	+10	+10	-5	159.16	+18.73



Discussion

- Technical feasibility confirmed (high capacity factor)
- Challenges: high CAPEX, grid bottlenecks, variability
- LCOE above North Sea benchmarks; scope to reduce via CAPEX/AEP optimisation

STAKEHOLDERS INTEREST & CONFLICTS

SOCIETAL EMBEDDING

Government

Climate targets, security of supply vs permitting delays

STAKEHOLDERS INTEREST & CONFLICTS

Government

Climate targets, security of supply vs permitting delays

INDUSTRY

Profitability vs policy/price uncertainty

STAKEHOLDERS INTEREST & CONFLICTS

STAKEHOLDERS INTEREST & CONFLICTS

Government

Climate targets, security of supply vs permitting delays

INDUSTRY

Profitability vs policy/price uncertainty

Local communities

Jobs, revenue sharing vs visual impact

STAKEHOLDE INTEREST & CONFLICTS

Government

Climate targets, security of supply vs permitting delays

INDUSTRY

Profitability vs policy/price uncertainty

Local communities

Jobs, revenue sharing vs visual impact

GRID

Small grid, slow changes

Government	INDUSTRY	Local communities	GRID	NGOs
Climate targets, security of supply vs permitting delays	Profitability vs policy/price uncertainty	Jobs, revenue sharing vs visual impact	Small grid, slow changes	Biodiversity & habitat protection

Government	INDUSTRY	Local communities	GRID	NGOs
HEREMA, OWFODA's, centralisation leases	Profitability vs policy/price uncertainty	Jobs, revenue sharing vs visual impact	Small grid, slow changes	Biodiversity & habitat protection

Government

HEREMA, OWFODA's, centralisation leases

INDUSTRY

Tax advantages
Large investments hubs

Local communities

Jobs, revenue sharing vs visual impact

GRID

Small grid, slow changes

NGOs

Biodiversity & habitat protection

Government

HEREMA, OWFODA's, centralisation leases

INDUSTRY

Tax advantages
Large investments hubs

Local communities

Social involvments, clear projects, benefits

GRID

Small grid, slow changes

NGOs

Biodiversity & habitat protection

Government

HEREMA, OWFODA's, centralisation leases

INDUSTRY

Tax advantages
Large investments hubs

Local communities

Social involvments, clear projects, benefits

GRID

Grid upgrades
Substations integration

NGOs

Biodiversity & habitat protection

Government

HEREMA, OWFODA's, centralisation leases

INDUSTRY

Tax advantages
Large investments hubs

Local communities

Social involvments, clear projects, benefits

GRID

Grid upgrades
Substations integration

NGOs

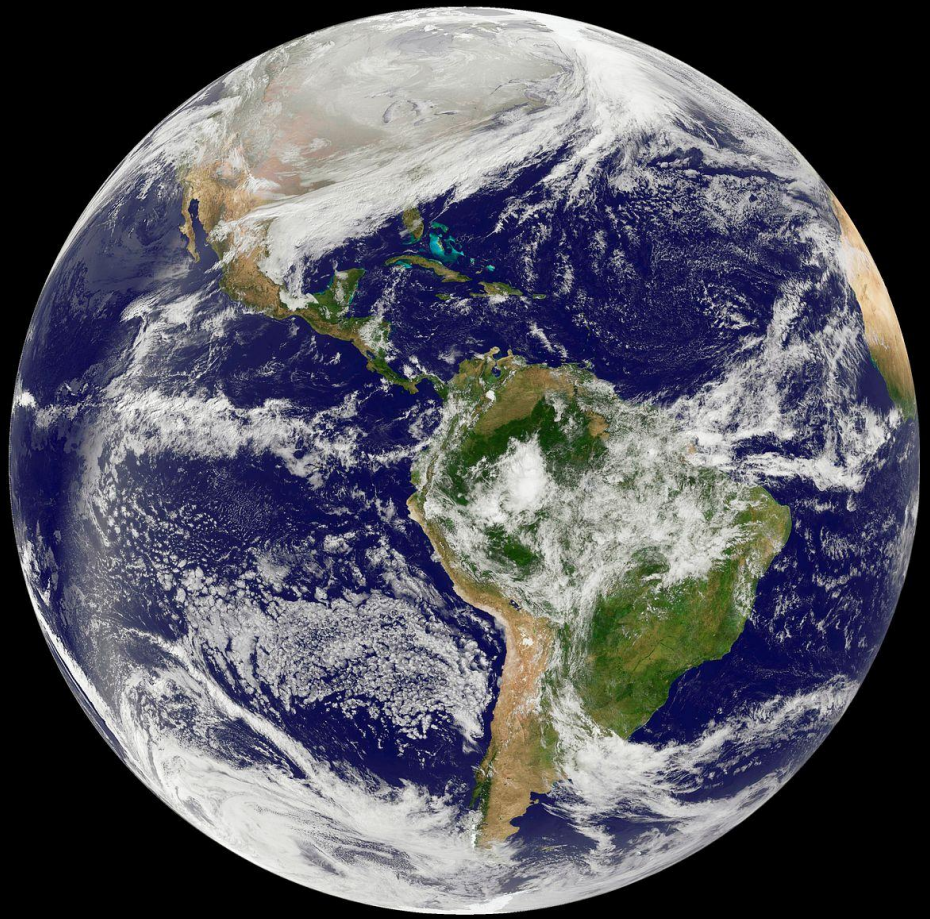
Species protection
Artificial reefs

Conclusion

- Offshore wind in Greece is technically feasible & profitable
- Strong NPV & IRR; CAPEX optimization is key for competitive LCOE
- Stakeholder management will determine deployment success

References

| **[1]** Q. A. Al-Haija, O. Mohamed, and W. A. Elhaija, “Predicting global energy demand for the next decade: A time-series model using nonlinear autoregressive neural networks,” *Energy Exploration & Exploitation*, vol. 41, no. 6, pp. 1884–1898, 2023.
| **[2]** D. Gielen, F. Boshell, D. Saygin, M. D. Bazilian, N. Wagner, and R. Gorini, “The role of renewable energy in the global energy transformation,” *Energy Strategy Reviews*, vol. 24, pp. 38–50, 2019.
|**[3]** I. Sohn, “Review: Global energy outlook 2024: Peaks or plateaus?,” *Foresight: The International Journal of Applied Forecasting*, pp. 44–46, 2024.
| **[4]** A. M. Jadali, A. Ioannou, K. Salonitis, *et al.*, “Decommissioning vs. repowering of offshore wind farms—a techno-economic assessment,” *Int. J. Adv. Manuf. Technol.*, vol. 112, pp. 2519–2532, Feb. 2021.
| **[5]** Copernicus Climate Change Service (C3S), “ERA5 hourly data on single levels,” 2018. [Online]. Available: https://cds.climate.copernicus.eu
| **[6]** S. A. Hsu, E. A. Meindl, and D. B. Gilhousen, “Determining the power-law wind-profile exponent under near-neutral stability at sea,” *J. Appl. Meteor. Climatol.*, vol. 33, no. 6, pp. 757–765, 1994.
| **[7]** N. Jensen, *A note on wind generator interaction*. Risø National Lab., Rep. 2411, 1983.
| **[8]** I. Katić, J. Højstrup, and N. O. Jensen, “A simple model for cluster efficiency,” in *Proc. Eur. Wind Energy Conf.*, 1986. |
| **[9]** WindEurope, “European offshore wind farms map (public),” 2025.
| **[10]** European Commission, “EU strategy for offshore renewable energy,” 2020.
| **[11]** Eurostat, “Renewable energy statistics,” 2025.
| **[12]** E. Faraggiana *et al.*, “Floating offshore wind potential for Mediterranean countries,” *Heliyon*, vol. 10, no. 13, p. e33948, 2024.
| **[13]** J. Schallenberg-Rodríguez and N. García Montesdeoca, “Spatial planning to estimate offshore wind potential: The Canary Islands,” *Energy*, vol. 143, pp. 91–103, 2018.
| **[14]** A. Kuzior *et al.*, “Assessment of the energy security of EU countries,” *Energies*, vol. 18, no. 8, 2025.
| **[15]** Hellenic Parliament, “Law 4964/2022, Articles 137–162,” 2022.
| **[16]** RAAEY, “National report 2023: Electricity & gas market in Greece,” Athens, Greece, Jun. 2023.
| **[17]** U.S. Int. Trade Administration, “Greece renewable energy projects 2024,” 2024.
| **[18]** S. Spyridonidou, D. G. Vagiona, and E. Loukogeorgaki, “Strategic planning of offshore wind farms in Greece,” *Sustainability*, vol. 12, no. 3, 2020.
[19] A. J. Schutte, “Assessment of technical feasibility for offshore wind turbines in the Aegean,” in *Proc. BSc Conf. 2025*, Univ. Twente, 2025.
| **[20]** Vestas Wind Systems A/S, “V236-15.0 MW™ Offshore Wind Turbine,” 2025. |
| **[21]** G. *et al.*, “Definition of the IEA 15-MW offshore reference wind turbine,” NREL, Rep. TP-5000-75698, Mar. 2020. \
| **[22]** T. W. May, Y. M. Yeap, and A. Ukil, “Comparative evaluation of HVAC and HVDC transmission,” in *TENCON 2016 – IEEE Region 10 Conf.*, Nov. 2016.
| **[23]** C. L. Archer *et al.*, “Quantifying array losses due to spacing in offshore wind farms,” in *AGU Fall Meeting Abstracts*, 2013.
| **[24]** The MathWorks Inc., *MATLAB R2024b*, 2024. |
| **[25]** A. Al-Quraan *et al.*, “Power curve modelling of wind turbines,” *IET Renew. Power Gener.*, vol. 16, no. 3, pp. 362–374, 2021.
| **[26]** A. Niayifar and F. Porté-Agel, “Analytical modeling of wind farms,” *Energies*, vol. 9, no. 9, 2016.
| **[27]** S. T. Frandsen, “Turbulence and structural loading in turbine clusters,” Forskningscenter Risø, Denmark, Rep. 1188, 2007.
| **[28]** T. H. Malik and C. Bak, “Challenges in detecting wind turbine power loss,” *Wind Energy Sci.*, vol. 10, no. 1, pp. 227–243, 2025. |
| **[29]** Y. Ma, C. L. Archer, and A. Vasel-Be-Hagh, “The Jensen wind farm parameterization,” *Wind Energy Sci.*, vol. 7, no. 6, pp. 2407–2431, 2022.
| **[30]** J. Carroll *et al.*, “Availability improvements from condition monitoring systems,” in *EWEA Offshore 2015*, Mar. 2015.
| **[31]** P. Laskowicz, “Offshore wind energy potential in Europe,” 2022.
| **[32]** N. Yildiz *et al.*, “O&M and decommissioning cost of floating turbines,” *Energies*, vol. 17, no. 6, p. 1332, 2024. |
| **[33]** A. Roth *et al.*, *Renewable energy financing conditions in Europe: Survey and impact analysis*. Aures, 2021.
| **[34]** *Guide to an Offshore Wind Farm*, “Wind farm costs,” 2025.
| **[35]** ENTSO-E Transparency Platform, “Installed capacity per production type: Greece,” 2025.
| **[36]** S. Cevik and K. Ninomiya, “Energy transition and electricity prices in Europe,” IMF Working Paper WP/22/220, Nov. 2022. |
| **[37]** Trading Economics, “Greece consumer price index (CPI),” 2025.



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